

Problem Set 5

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1 Combinatorics

For each question below please write a brief paragraph explaining your reasoning, *it is not sufficient to only write the answer*.

Problem 1: How many ways can 7 players be chosen from a group of 11 possible players if:

1. There is no restriction on the selection.
2. A fixed player needs to be on the team.
3. A fixed player is never chosen.

Problem 2: How many ways can you choose 4 groups of 4 people from 20 people, assuming the groups are distinct?

Problem 3: How many distinct permutations are there of letters in the word “Illinois”? In your answer, ignore issues of case (i.e., let I be equivalent to i).

Problem 4: How many triangles can be formed by 9 points of which 3 are colinear (and all the rest of the points are *not colinear*)?